MAT 3007 – Optimization

Branch and Bound

Lecture 28

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CUHK-Shenzhen

Recap

LP Relaxation

Integer Problems:

$$\text{minimize}_{\boldsymbol{x}} \ \boldsymbol{c}^\top \boldsymbol{x} \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}, \quad \boldsymbol{x} \in \mathbb{Z}^n.$$

LP Relaxation:

- Relax the integer constraints and solve a linear program.
- For minimization problems, the optimal value of the LP relaxation provides a ~~upper~~ lower bound for the optimal value of IP.
- For minimization problems, the value of a feasible integer point provides a ~~lower~~ upper bound for the optimal value of IP.
- If the optimal solution of the LP relaxation is an integer point, then is also optimal to the IP.

Total Unimodularity: If the constraint matrix $\boldsymbol{A}$ is TU and $\boldsymbol{b}$ is an integer vector, then:

- All basic feasible solutions are integers.
- Solving the LP via the simplex method will solve the IP.

Total Unimodularity

We introduce a condition under which all BFS must be integers — total unimodularity.

Definition: Total Unimodularity

A matrix $\boldsymbol{A}$ is said to be totally unimodular if the determinant of each submatrix of $\boldsymbol{A}$ is either 0, 1, or -1.

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Theorem: Total Unimodularity and Integer Solutions

If the constraint matrix $\boldsymbol{A}$ is totally unimodular and $\boldsymbol{b}$ is an integer vector, then all the BFS for the LP relaxation are integers. If it has an optimal solution, then it must have an optimal solution that is an integer solution.

$\longrightarrow$ we need an algorithm that can find optimal BFS, e.g. simplex method.

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Total Unimodularity (TU):

- In order for $\boldsymbol{A}$ to be TU, it can only have 0, 1 or -1 entries.
- But it requires much more...

One Sufficient Condition for TU

Theorem

Let $\boldsymbol{A}$ be an $m \times n$ matrix. Then the following conditions together are sufficient for $\boldsymbol{A}$ to be totally unimodular:

1. Every column of $\boldsymbol{A}$ contains at most two non-zero entries;
2. Every entry in $\boldsymbol{A}$ is 0, $+1$, or -1;
3. The rows of $\boldsymbol{A}$ can be partitioned into two disjoint sets B and C such that:
   - If two non-zero entries in a column of $\boldsymbol{A}$ have the same sign, then the row of one entry is in B, and the row of the other in C;
   - If two non-zero entries in a column of $\boldsymbol{A}$ have opposite signs, then the rows are both in B, or both in C.

Example: Matching Problem

Recall the matching problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n} \sum_{j=1}^{n} v_{ij} x_{ij} & \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_{ij} = 1 & \forall\ j \\ & \sum_{j=1}^{n} x_{ij} = 1 & \forall\ i \\ & x_{ij} \in \{0, 1\} & \end{aligned}$$

The constraint matrix is TU.

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An example when $n = 2$ (in the order of $x_{11}, x_{12}, x_{21}, x_{22}$):

$$\boldsymbol{A} = \begin{array}{c} \\ B \\ B \\ C \\ C \end{array} \begin{array}{c} \begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

Choosing $B = \{1, 2\}$, $C = \{3, 4\}$, the condition holds and $\boldsymbol{A}$ is TU.

Branch-and-Bound Method

The General Process – Branching

First, we solve the LP relaxation of the IP:

- If the solution is an integer, then it is optimal to the IP.

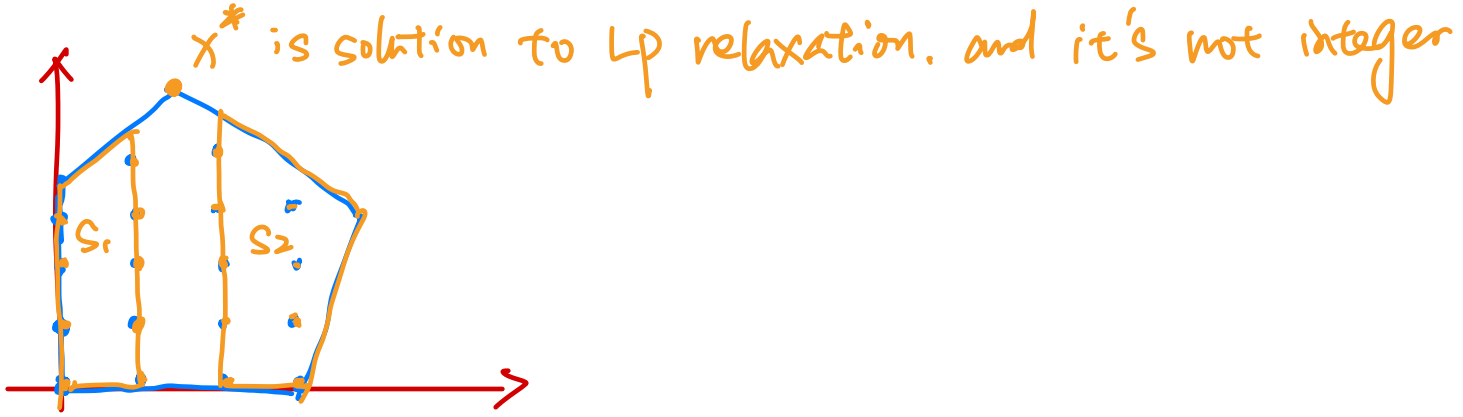
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If the optimal solution to the LP relaxation is $\boldsymbol{x}^*$ and $x_i^* \notin \mathbb{Z}$, then branch the problem into the following two:

1. One with an added constraint $x_i \leq \lfloor x_i^* \rfloor$, we call this (S1).
2. One with an added constraint $x_i \geq \lceil x_i^* \rceil$, we call this (S2).



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Procedure: (for max problems)

- We solve (S1) and (S2) and assume we get the solutions $\boldsymbol{y}_1^*$ and $\boldsymbol{y}_2^*$ with optimal values v_1^* and v_2^* ((S1) and (S2) are IPs).
- If $v_2^* \leq v_1^*$, then $\boldsymbol{y}_1^*$ is the optimal solution to the original IP. Otherwise, $\boldsymbol{y}_2^*$ is the optimal solution.

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- If $v_2^* \leq v_1^*$, then $\boldsymbol{y}_1^*$ is the optimal solution to the original IP. Otherwise, $\boldsymbol{y}_2^*$ is the optimal solution.

Question: How to solve (S1) and (S2)?

$\rightsquigarrow$ Use the same idea (solve LP relaxation and further branch).

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- **Upper bound:** The LP relaxation solution will be an upper bound — the objective value of any integer solution from this node must be no greater than than the optimal value of the relaxed LP.

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Bounding

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- **Upper bound:** The LP relaxation solution will be an upper bound — the objective value of any integer solution from this node must be no greater than than the optimal value of the relaxed LP.
- **Lower bound:** The objective value of any feasible (integer) point is a lower bound for the optimal value — the optimal solution of the IP must be no less than the objective value achieved by any feasible point.

Bounding: Pruning the Branches

Bounding Procedure:

- At a certain node, when the optimal value of the LP relaxation of this branch is even less than the current lower bound, then we can abandon this branch.

or no better than

Bounding: Pruning the Branches

Bounding Procedure:

- At a certain node, when the optimal value of the LP relaxation of this branch is even less than the current lower bound, then we can abandon this branch.
- These results will be the opposite if we are minimizing.

We will use the bounding steps to prune unnecessary computations (or in other words, remove unnecessary branches).

Example

Consider the earlier example

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0, \quad x_1, x_2 \in \mathbb{Z} \end{aligned}$$

- We solve the LP relaxation and get $\boldsymbol{x}^* = (15/4, 9/4)$.
- We first branch for x_1.

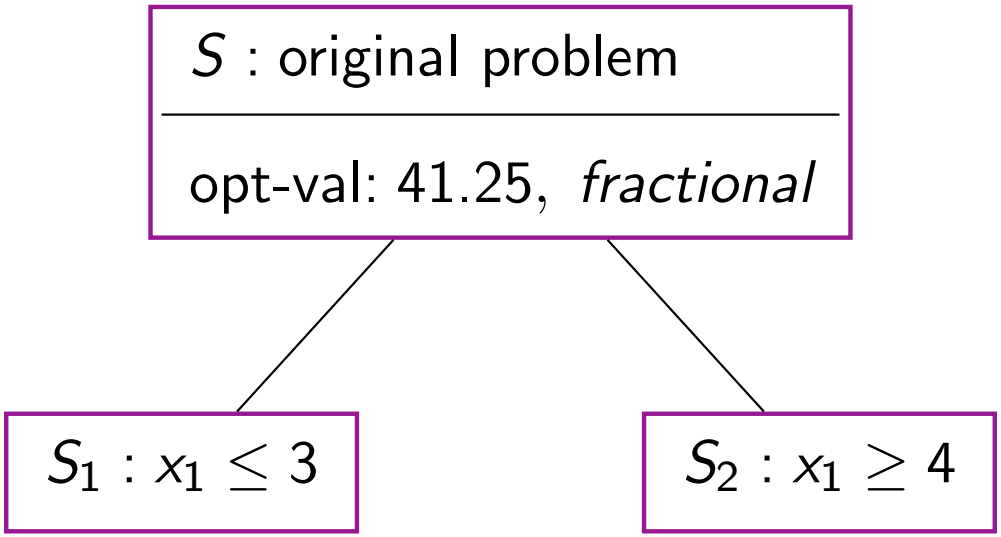
We solve two subproblems:

- One with an additional constraint $x_1 \leq 3 \rightsquigarrow$ problem (S1).
- One with an additional constraint $x_1 \geq 4 \rightsquigarrow$ problem (S2).

$\lfloor 15/4 \rfloor$ ↙

↖ $\lceil 15/4 \rceil$

Example: Continued



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We solve the LP relaxation of (S1) first:

- The optimal solution is $(3, 3)$. This is an integer solution, so we are done with this branch (the optimal value is 39).

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Then we solve the LP relaxation of (S2):

- The optimal solution is $(4, 1.8)$ (the optimal value is 41).

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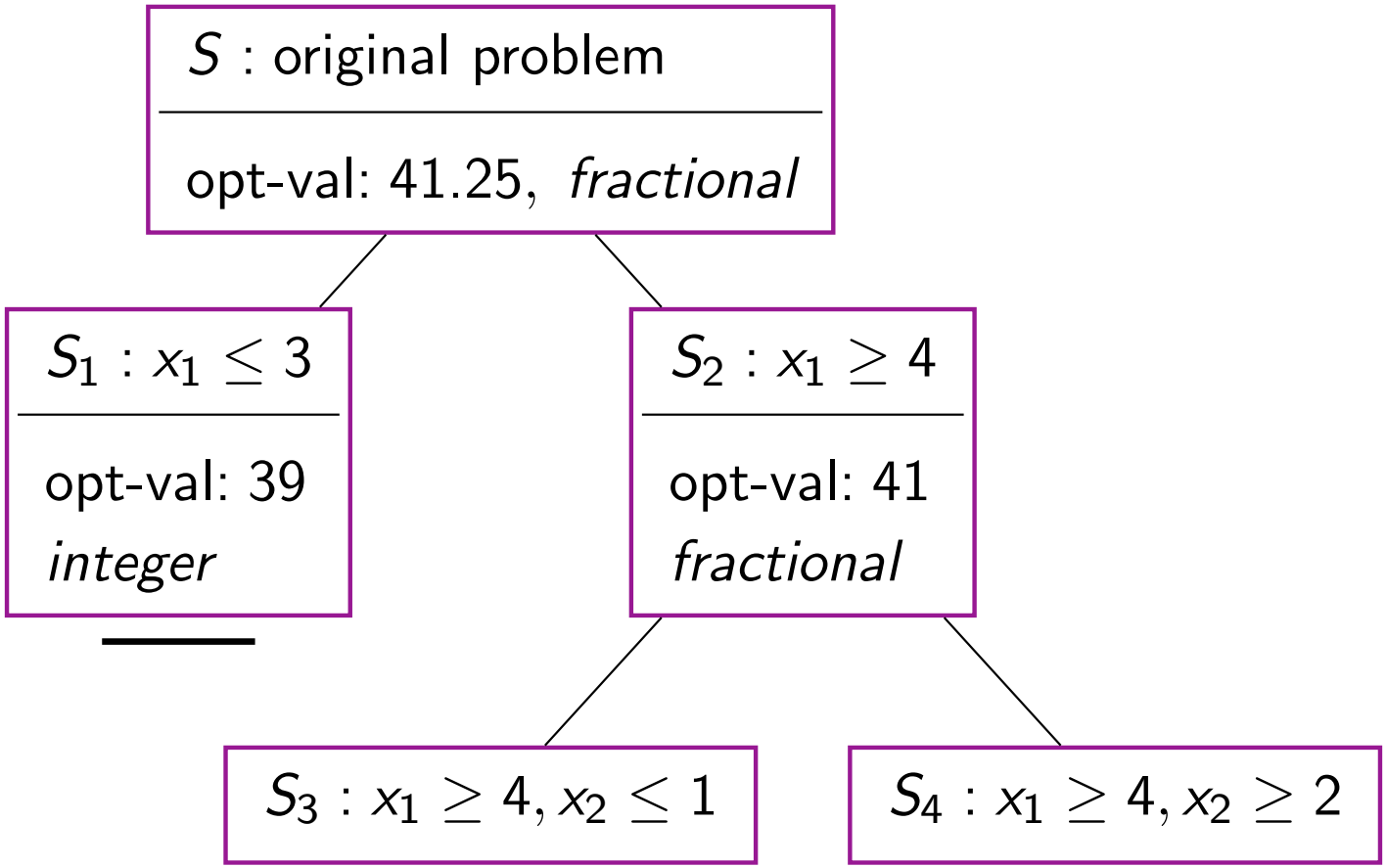
Then we solve the LP relaxation of (S2):

- The optimal solution is $(4, 1.8)$ (the optimal value is 41).

The solution to (S2) is not an integer. We have to do further branching:

- We add a constraint $x_2 \leq 1 \rightsquigarrow$ (S3).
- We add a constraint $x_2 \geq 2 \rightsquigarrow$ (S4).

Example: Continued



Solving (S3)

Subproblem (S3):

$$
\begin{aligned}
\text{maximize} \quad & 8x_1 + 5x_2 \\
\text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\
& x_1 + x_2 \leq 6 \\
& x_1 \geq 4, x_2 \leq 1 \quad x_1, x_2 \geq 0, \quad x_1, x_2 \in \mathbb{Z}
\end{aligned}
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The optimal solution of LP relaxation is $(40/9, 1)$, we further branch:

$\swarrow \lfloor 40/9 \rfloor$

- Add a constraint $x_1 \leq 4 \rightsquigarrow$ (S5).
- Add a constraint $x_1 \geq 5 \rightsquigarrow$ (S6).

$\uparrow \lceil 40/9 \rceil$

Solving (S3)

Subproblem (S3):

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1 \geq 4, x_2 \leq 1 \quad x_1, x_2 \geq 0, \quad x_1, x_2 \in \mathbb{Z} \end{aligned}$$

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- Add a constraint $x_1 \leq 4 \rightsquigarrow$ (S5).
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For (S5), x_1 has to be 4 and the optimal solution is $(4, 1)$ with objective value 37 (already integer, so do not need to do further branching).

Subproblem (S3):

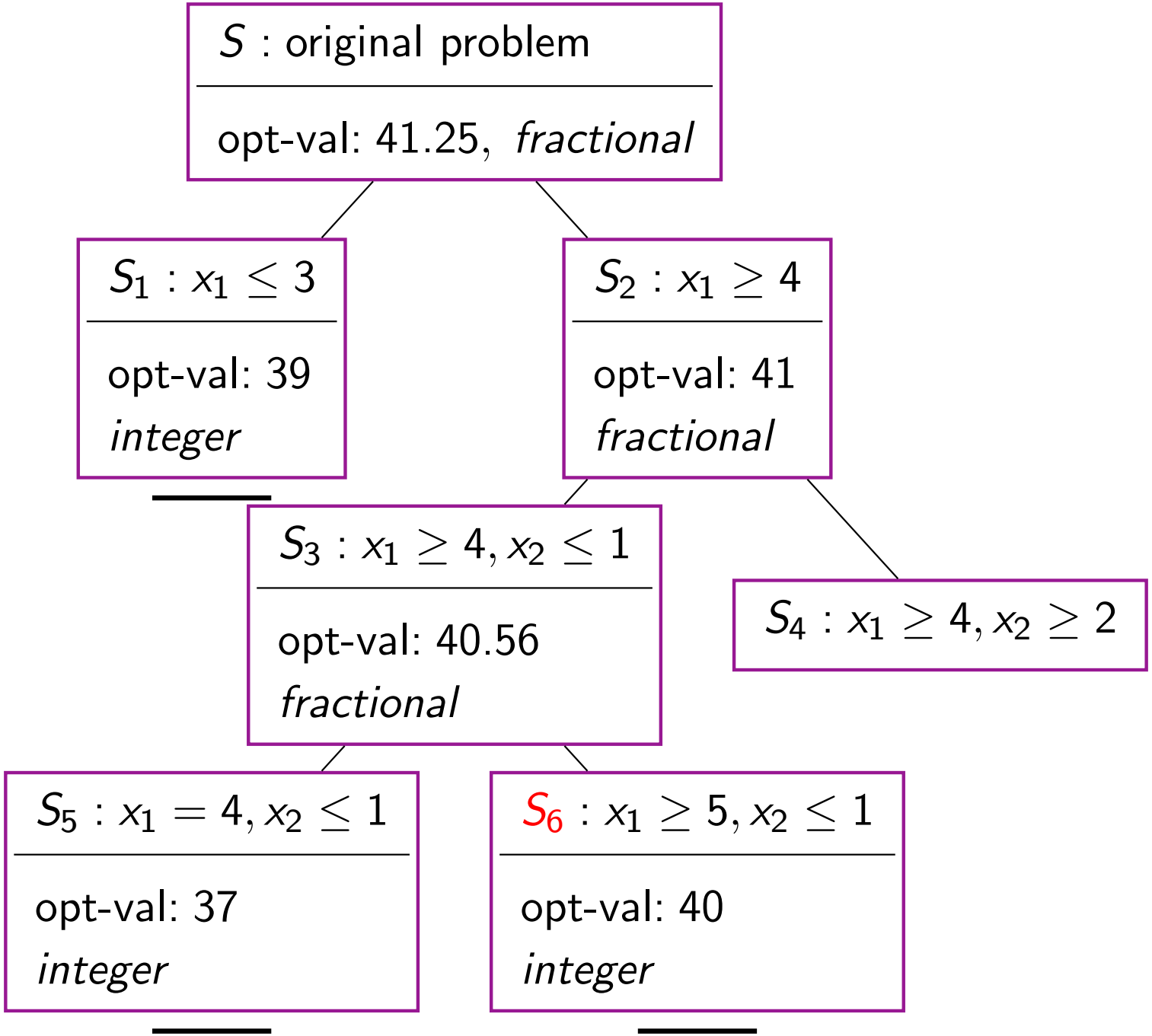
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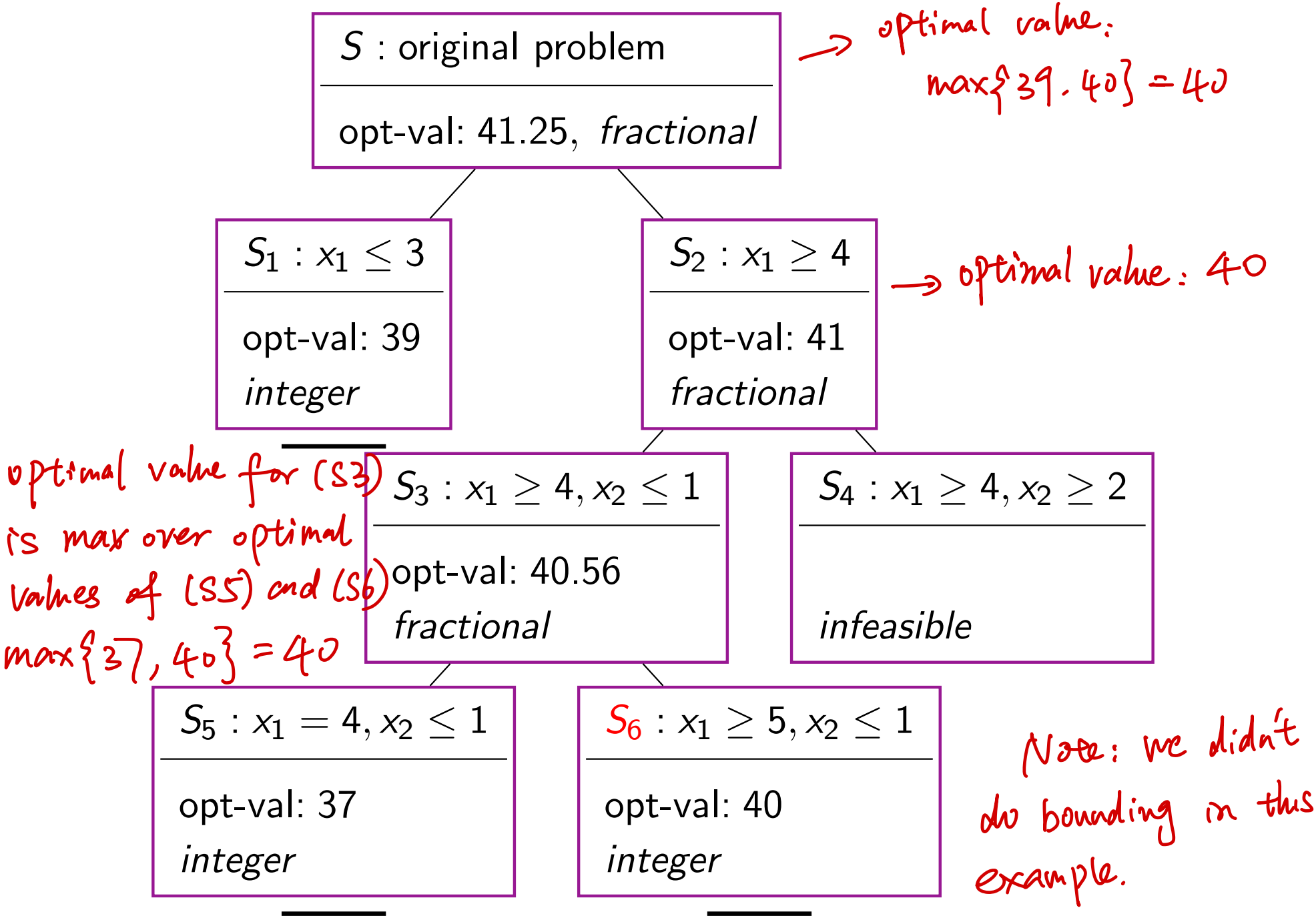
For (S6), the optimal solution is $(5, 0)$, the objective value is 40.


S : original problem
opt-val: 41.25, *fractional*
$S_1 : x_1 \leq 3$
opt-val: 39
integer
$S_2 : x_1 \geq 4$
opt-val: 41
fractional
$S_3 : x_1 \geq 4, x_2 \leq 1$
opt-val: 40.56
fractional
$S_4 : x_1 \geq 4, x_2 \geq 2$
$S_5 : x_1 = 4, x_2 \leq 1$
opt-val: 37
integer
$S_6 : x_1 \geq 5, x_2 \leq 1$
opt-val: 40
integer

Subproblem (S4):

$$
\begin{aligned}
\text{maximize} \quad & 8x_1 + 5x_2 \\
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& x_1 + x_2 \leq 6 \\
& x_1 \geq 4, x_2 \geq 2 \qquad x_1, x_2 \geq 0, \qquad x_1, x_2 \in \mathbb{Z}
\end{aligned}
$$

One can easily see that (S4) is not feasible. Thus we do not need to further consider this subproblem.


S : original problem
opt-val: 41.25, *fractional*
optimal value:
$\max\{39, 40\} = 40$
$S_1 : x_1 \leq 3$
opt-val: 39
integer
$S_2 : x_1 \geq 4$
opt-val: 41
fractional
optimal value: 40
optimal value for (S3) is max over optimal values of (S5) and (S6)
$\max\{37, 40\} = 40$
$S_3 : x_1 \geq 4, x_2 \leq 1$
opt-val: 40.56
fractional
$S_4 : x_1 \geq 4, x_2 \geq 2$
infeasible
$S_5 : x_1 = 4, x_2 \leq 1$
opt-val: 37
integer
$S_6 : x_1 \geq 5, x_2 \leq 1$
opt-val: 40
integer
Note: we didn't do bounding in this example.

Branch-and-Bound Method: Summary

High-Level Idea:

- **Branching:** Divide the feasible region into smaller ones, solve each of them and combine them to find the optimal solution.
- **Bounding:** Use bounds (LP optimal value and feasible points) to reduce the number of branches we need to consider.

Order of Computing the Branches

When we perform branch-and-bound, we may have two choices at each step:

- In the example, when we compute (S3), then we get a non-integer solution and thus two branches.
- Then we need to decide if we want to continue with one of the new branches of (S3) or try (S4) first.

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When we perform branch-and-bound, we may have two choices at each step:

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Basically, we need to decide if we want to go deep into one branch first or go wide to solve all problems on a given level.

Deep or Wide?

In the branch-and-bound algorithm, the best approach is to go deep into the tree, not to go wide:

Deep or Wide?

In the branch-and-bound algorithm, the best approach is to go deep into the tree, not to go wide:

- Most integer solutions lie deep in the tree. It is good to have integer feasible solutions early, so we can use it in the bounding procedure.
- It is also memory-efficient, since each LP is obtained from its parent by merely adding one constraint.
- It is also easier to code (recursion).

Complexity of Branch-and-Bound

Branch-and-bound is essentially an enumeration method.

- In the worst case, branch-and-bound may need to go through each feasible integer point in the region, which is exponential in the problem size.
- Remember there is no polynomial-time algorithm for IP.

However, branch-and-bound does enumeration in a smart way and typically it only needs to visit a tiny fraction of all solutions.

- Much more efficient than explicitly enumerating the solutions.
- It is one of the most useful practical methods.

Branch-and-Bound for Binary Problems

Branch-and-Bound for Binary Problem

Now we use Branch-and-bound to solve binary linear programs.

Consider the following knapsack problem:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_j \in \{0, 1\}, \quad j = 1, \ldots, 4. \end{aligned}$$

Also check out last problem of the sample final.

Example Continued

The LP relaxation for this IP is:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & 0 \leq x_j \leq 1, \quad j = 1, \ldots, 4. \end{aligned}$$

replace $x_j \in \{0, 1\}$ by

Example Continued

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In this case, the optimal solution is $x_1 = 1$, $x_2 = 1$, $x_3 = 0.5$, $x_4 = 0$. The optimal value is 22.

Example Continued

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- We need to do branching for x_3.
- Consider two subproblems, one with $x_3 = 1$ ($\rightsquigarrow$ (S1)), the other with $x_3 = 0$ ($\rightsquigarrow$ (S2)).

Example Continued

Solving the LP relaxation of (S1),

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_3 = 1, \quad x_1, x_2, x_4 \in [0, 1]. \end{aligned}$$

The optimal solution $x_1 = 1$, $x_2 = 0.714$, $x_3 = 1$ and $x_4 = 0$. The optimal value of the LP relaxation is 21.85.

Example Continued

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- Still fractional, so we further branch.
  - $x_2 = 0 \rightsquigarrow$ (S3).
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- Still fractional, so we further branch.
  - $x_2 = 0 \rightsquigarrow$ (S3).
  - $x_2 = 1 \rightsquigarrow$ (S4).
- We obtained an important information: The optimal value of (S1) can not be better than 21.85.

upper bound for the current subproblem

Example Continued

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  - $x_2 = 0 \rightsquigarrow$ (S3).
  - $x_2 = 1 \rightsquigarrow$ (S4).
- We obtained an important information: The optimal value of (S1) can not be better than 21.85.
- Since the optimal value of (S1) must be an integer, it is at most 21. because x_i are integers and the coeffients in the obj function are integers.

Example Continued

Subproblem (S3): Consider the LP relaxation:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_3 = 1, \quad x_2 = 0, \quad x_1, x_4 \in [0, 1]. \end{aligned}$$

The optimal solution is $x_1 = 1$, $x_2 = 0$, $x_3 = 1$, $x_4 = 1$. This is an integer solution with optimal value 18. Therefore, we are done with this branch.

This optimal sol. to the subproblem should be a feasible point to the original IP $\longrightarrow$ we get a lower bound

Example Continued

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The optimal solution is $x_1 = 1$, $x_2 = 0$, $x_3 = 1$, $x_4 = 1$. This is an integer solution with optimal value 18. Therefore, we are done with this branch.

- A lower bound of 18 is obtained. The optimal value of the original IP is at least 18.
- If we solve a later LP relaxation and the optimal value is less than 18, then we don't need to further consider that branch.

Example Continued

Subproblem (S4): Consider the LP relaxation:

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The optimal solution is $x_1 = 0.6$, $x_2 = 1$, $x_3 = 1$, $x_4 = 0$ and the optimal value is 21.8.

Example Continued

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- Still fractional. We need to do further branching.
  - $x_1 = 1 \rightsquigarrow$ (S5).
  - $x_1 = 0 \rightsquigarrow$ (S6).

Example Continued

Consider subproblem (S5):

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_3 = 1, \quad x_2 = 1, \quad x_1 = 1, \quad x_4 \in \{0, 1\}. \end{aligned}$$

It is easy to see that (S5) is infeasible. We do not need to further consider it.

Example Continued

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Consider (S6):

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_3 = 1, \quad x_2 = 1, \quad x_1 = 0, \quad x_4 \in \{0, 1\}. \end{aligned}$$

we update lower bound from 18 to 21.

The optimal solution is $x_1 = 0$, $x_2 = 1$, $x_3 = 1$ and $x_4 = 1$. The optimal value is 21, and it is a lower bound for IP.

Example Continued

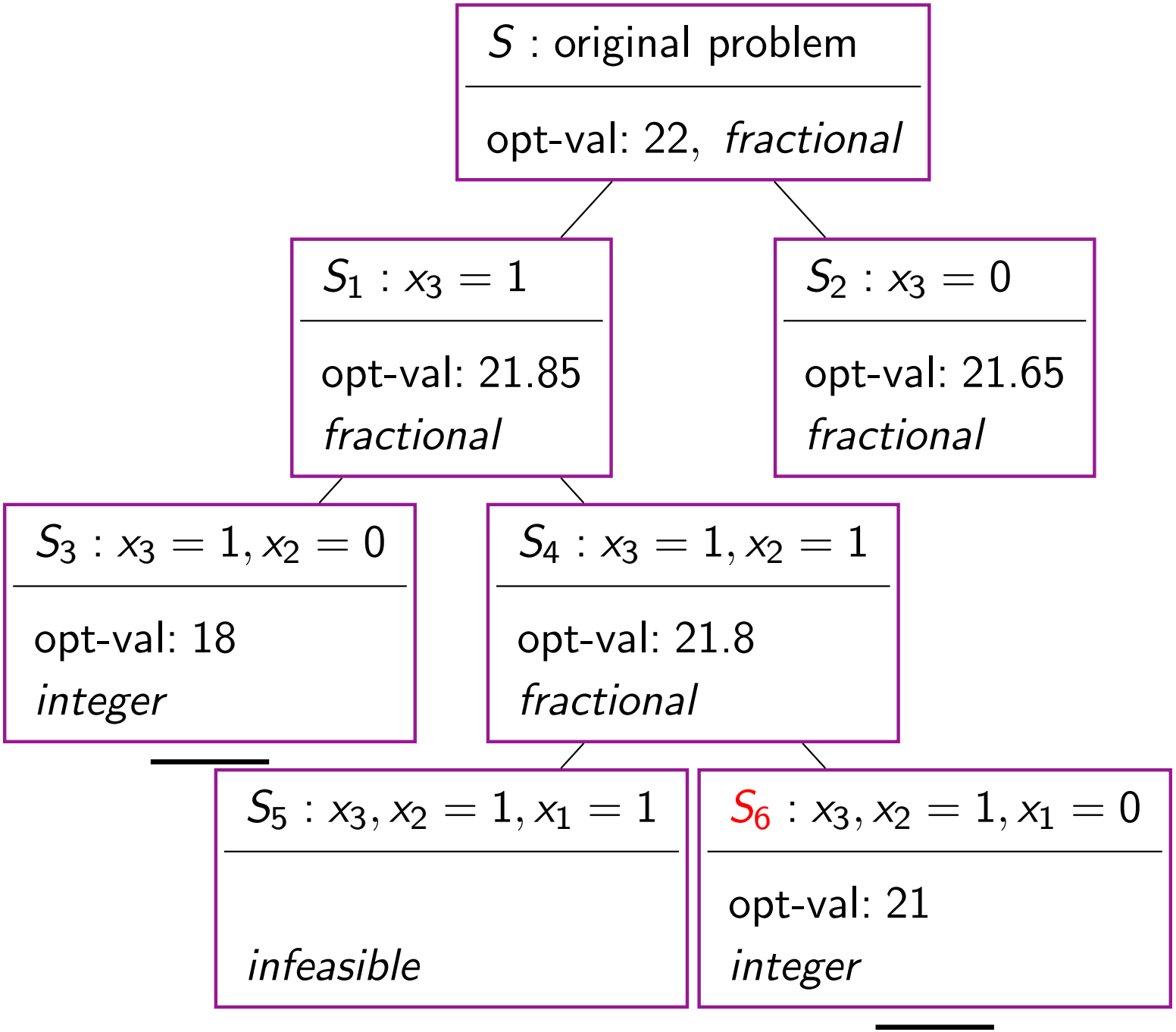
Therefore, we get 21 as the optimal value for the first branch (S1) with optimal solution $\boldsymbol{y}^* = (0, 1, 1, 1)^\top$.

Now consider (S2) and the LP relaxation:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_3 = 0, \quad x_1, x_2, x_4 \in [0, 1]. \end{aligned}$$

The optimal value is 21.65.

- 21.65 is an upper bound on this branch.
- Since the optimal value of (S2) must be integer, it means it can not be larger than 21. this is no better than current lower
- Therefore, no better solution can be obtained in this branch. bound.
  We don't need to consider it.


S : original problem
opt-val: 22, *fractional*
$S_1 : x_3 = 1$
opt-val: 21.85
fractional
$S_2 : x_3 = 0$
opt-val: 21.65
fractional
$S_3 : x_3 = 1, x_2 = 0$
opt-val: 18
integer
$S_4 : x_3 = 1, x_2 = 1$
opt-val: 21.8
fractional
$S_5 : x_3, x_2 = 1, x_1 = 1$
infeasible
$S_6 : x_3, x_2 = 1, x_1 = 0$
opt-val: 21
integer

Example: Summary

There are 16 possible combinations in total, but we don't need to visit all of them.

- Bounding is very important, it can greatly reduce the search space.
- In the above example, we do not need to consider the $x_3 = 0$ branch because of bounding.

Solving Integer Optimization with Software

CVX can solve (mixed) integer programs with installed Gurobi solver (both are free for academic use):

- You can find the instructions online on the CVX website.
- You need to download Gurobi and obtain a Gurobi license.
- Install them properly (you need to follow the steps online).
- When using CVX, add the command "`cvx_solver gurobi`" at the top and the keyword "`integer`" or "`binary`" when declaring variables.

One can also use the MATLAB function: `intlinprog`.

Summary for IP

- Modeling
- LP Relaxation
  - An upper and lower bounds for original IP.
  - Ideal Case: the optimal solution of the LP relaxation is an integer point, then is also optimal to IP.
- Total Unimodularity
  - Definition and a sufficient condition for TU.
  - If $\boldsymbol{A}$ is TU and $\boldsymbol{b}$ is an integer vector, then all BFS for the LP relaxation are integers.
- Branch and Bound
  - Branching.
  - Bounding.

Frank-Wolfe Algorithm for Constrained Problems

Optional (not Covered in the Final)

Constrained Optimization

We consider the following constrained optimization problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \Omega, \end{aligned}$$

where Ω be a nonempty, closed, and convex set.

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Projected Gradient Descent:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_{\Omega}(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

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Projected Gradient Descent:

$$\boldsymbol{x}^{k+1} = \mathcal{P}_{\Omega}(\boldsymbol{x}^k - \lambda_k \nabla f(\boldsymbol{x}^k)).$$

- The (Euclidean) projection, $\mathcal{P}_{\Omega}(\boldsymbol{x})$, of $\boldsymbol{x}$ onto Ω is the solution to
$$\min_{\boldsymbol{y}} \ \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{y}\|^2 \quad \text{s.t.} \quad \boldsymbol{y} \in \Omega$$
- The projection onto certain special sets, such as box constraints or ball constraints, can be computed using explicit formulas. However, for more general sets, an algorithm is typically required to compute the projection.

Frank-Wolfe Algorithm: Iterative Process

Now we consider a new algorithm for constrained optimization.

Frank-Wolfe Algorithm:

- At iteration k, solve the following *linearized subproblem*:

$$\boldsymbol{s}^k = \arg\min_{\boldsymbol{s}\in\Omega} \nabla f(\boldsymbol{x}^k)^\top \boldsymbol{s}.$$

- Update the solution:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \lambda_k(\boldsymbol{s}^k - \boldsymbol{x}^k),$$

  where $\lambda_k \in [0, 1]$ is a step size.

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Step Size Selection:

- Fixed step size: $\lambda_k = \frac{2}{k+2}$ is commonly used.

decreases with k

Geometric Intuition:

- At each iteration, $\boldsymbol{s}^k$ can be considered as the solution to

$$\boldsymbol{s}^k = \arg\min_{\boldsymbol{s}\in\Omega} f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{s} - \boldsymbol{x}^k).$$

linear approx. to f at x^k.

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- $\boldsymbol{s}^{k+1}$ is a convex combination of the current point $\boldsymbol{x}^k$ and the solution $\boldsymbol{s}^k$ of the linearized subproblem, so it is still in Ω:

$$\boldsymbol{x}^{k+1} = (1-\lambda_k)\boldsymbol{x}^k + \lambda_k \boldsymbol{s}^k,$$

$\in\Omega$ $\in\Omega$

With $\lambda_k = \frac{2}{k+2}$, x^{k+1} is closer to last iterate as k increases.

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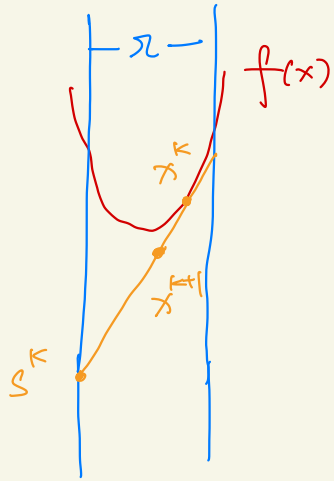
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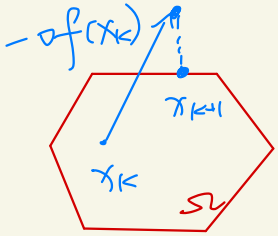
Comments:

- We still need to solve the linearized subproblem.
- Sometimes, the subproblem has a close-form solution and is easier to compute than projections. For example, $\{\boldsymbol{x} : \|\boldsymbol{x}\|_p \leq 1\}$.

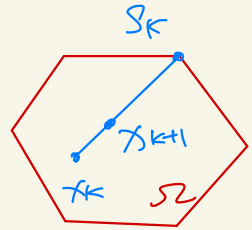
Frank-Wolfe :



Comprison with projected gradient descent



Proj GD



Frank Wolfe

Theorem: Convergence

Theorem: If f is convex and Lipschitz smooth with constant L, then the outputs of Frank-Wolfe algorithm with step size $\lambda_k = \frac{2}{k+2}$ satisfy:

$$f(\boldsymbol{x}^k) - f^* \leq \frac{2Ld_\Omega^2}{k+2},$$

where f^* is the optimal function value and $d_\Omega \triangleq \max_{\boldsymbol{x},\boldsymbol{y}\in\Omega} \|\boldsymbol{x}-\boldsymbol{y}\|$.

converges if Ω is bounded

$(d_\Omega < \infty)$

Today's lecture

- Branch-and-Bound
- Frank-Wolfe Algorithm

Next lecture

- Final Review Session

Questions?